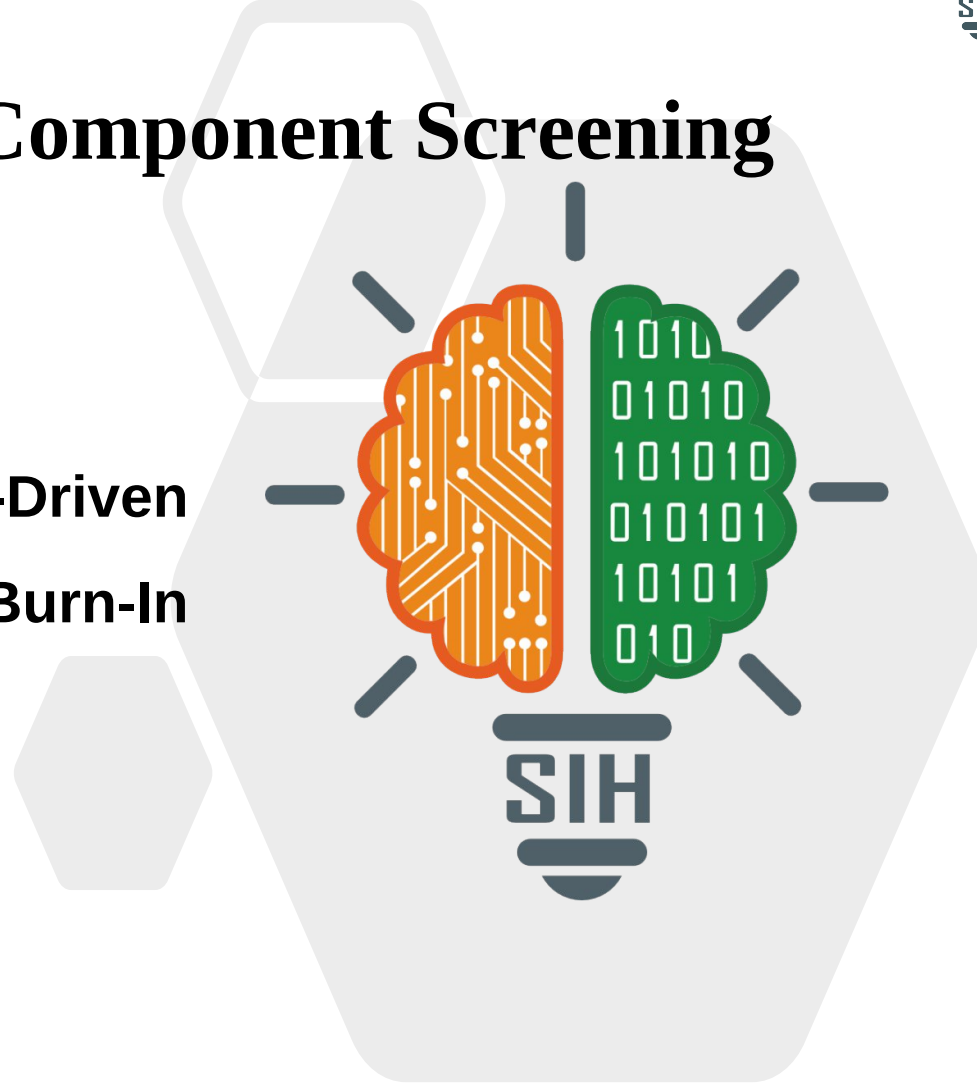


ACS - Adaptive Component Screening

- Problem Statement ID – 26170
- Problem Statement Title – AI-Driven Anomaly Detection in Component Burn-In & Screening
- Theme – Smart Automation
- PS Category – Software
- Team ID – EC004
- Team Name – Arch Rival



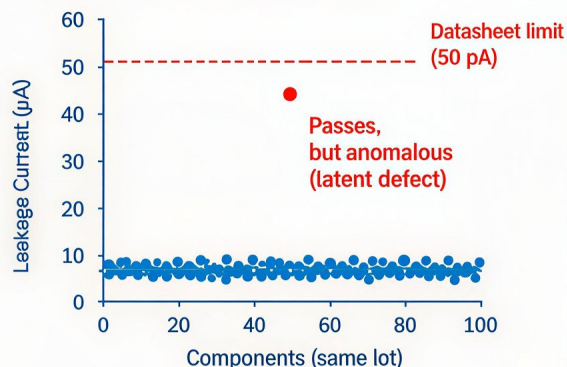
Adaptive AI Screening for Early Detection of Latent Electronic Component Failures

Traditional Screening (Existing Approach)

Rule-Based • Static • Parameter-wise

Each parameter is checked independently against a fixed datasheet limit. Components within the limit are considered PASS, even if they show abnormal trends or population-level deviations.

Parameter Check (Example: Iddq)



- Normal components
- Latent defective (still within limit)

Characteristics

- Fixed datasheet limits (per parameter)
- Single-time measurements
- No population or temporal context
- Ignores subtle multivariate patterns
- Cannot predict future drift
- Limited handling of out-of-distribution components

Key Limitations

- ❗ Misses latent defects within limits
- ❗ High false confidence
- ❗ No early warning
- ❗ No uncertainty estimation
- ❗ Not scalable to new component families

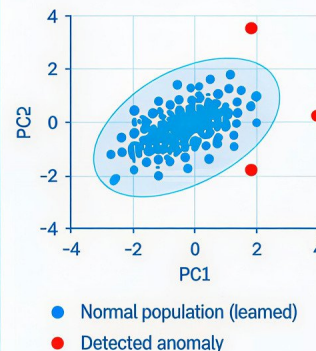
FROM
LIMIT
CHECKS
TO
LEARNING
BEHAVIOUR

ACS – Adaptive Component Screening (Our Approach)

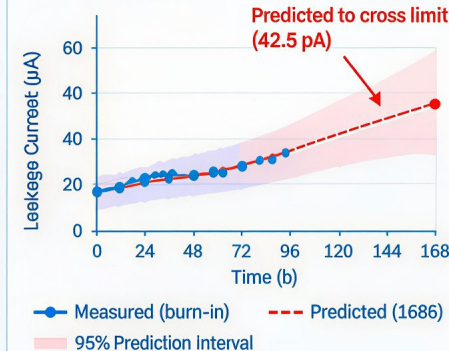
Data-Driven • Population-Aware • Predictive

Learns the normal behaviour from population and temporal data, detects subtle anomalies, forecasts future drift (168h) and provides an explainable, safety-gated engineering decision.

1. Population-Aware Detection (Multivariate)



2. Time-Series Drift Prediction (Example: Iddq)



3. Decision with Uncertainty

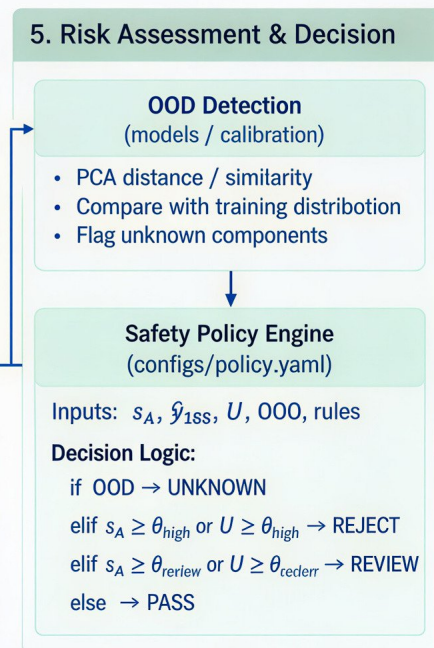
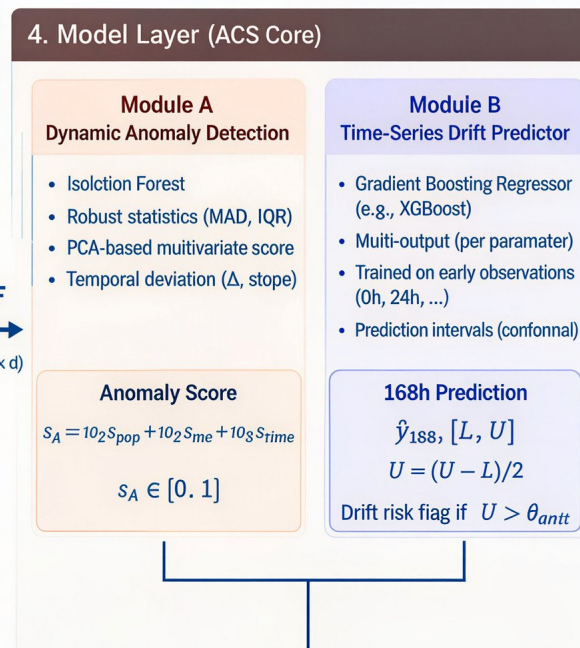
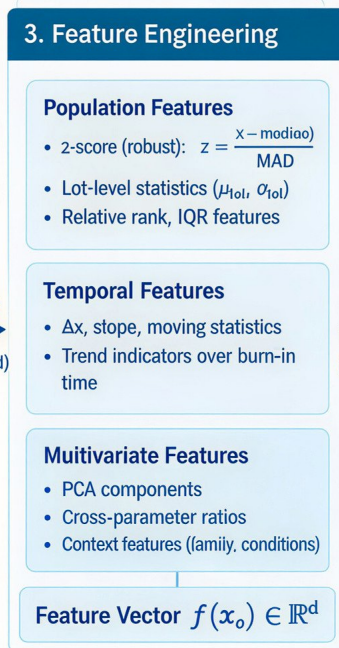
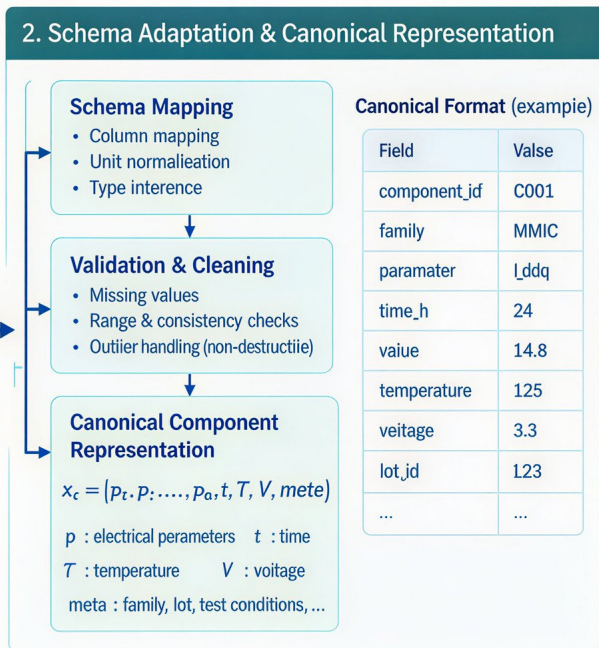
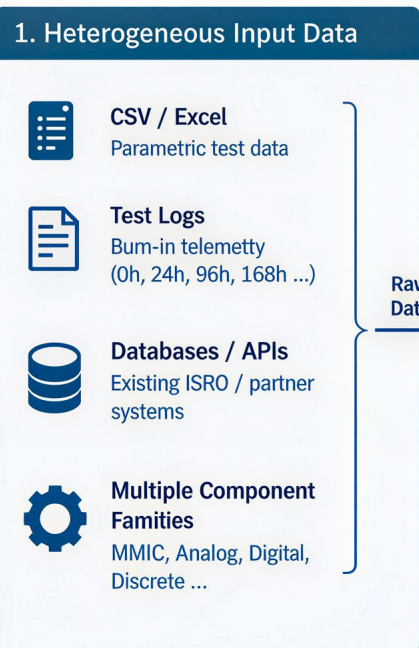
- PASS** Normal behaviour, low risk
- REVIEW** Borderline / needs engineering attention
- REJECT** High risk of future failure
- UNKNOWN** Out-of-distribution (unknown component)

Why It's Better

- ✓ Detects subtle, population-relative anomalies
- ✓ Uses temporal trends to forecast 168h behaviour
- ✓ Combines multiple parameters (multivariate)
- ✓ Handles out-of-distribution components
- ✓ Provides uncertainty with predictions
- ✓ Generates human-readable explanations
- ✓ Configurable safety policy aligned with mission requirements

Key Differences at a Glance

Decision Logic		Data Used		Anomaly Detection		Future Prediction		Uncertainty Handling		Explainability	
Fixed limits (per parameter)	→	Learned normal behaviour + predicted drift	→	Single-time measurement	→	Time-series + population + multivariate	→	Threshold-based	→	Isolation Forest + Robust statistics (MAD, IGR) + PCA-based score	→
								Not available	→	168h forecast (Gradient Boosting) with uncertainty (Conformal)	→
								None	→	Prediction intervals and OOD detection (PCA distance)	→
										Pass/Fail only	→
										Parameter-wise reasons, SHAP contributions, counterfactuals	



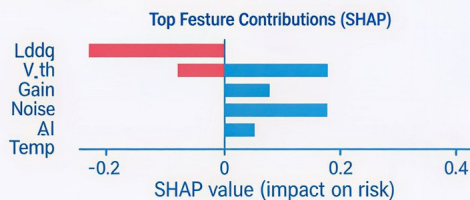
6. Outputs & Deliverables

Component Disposition



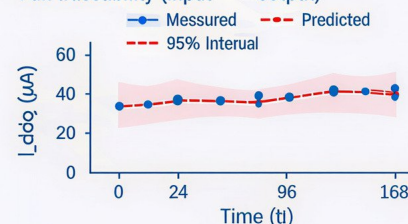
Explainability (Per Component)

- Top contributing parameters (SHAP)
- Direction of impact ($\uparrow$ / $\downarrow$)
- Counterfactual suggestion (what change could make it pass)
- Human-readable reason codes

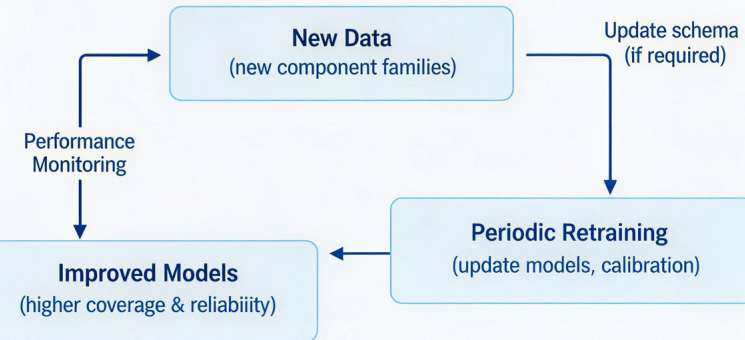


Engineering Report

- Parameter plots (measured vs. predicted with interval)
- Anomaly score breakdown
- Final decision with reasons
- Full traceability (input → output)



7. Feedback Loop



1. WORKING PROTOTYPE / IMPLEMENTED STACK

Runnable End-to-End Pipeline

Tech Stack (Actual Versions)

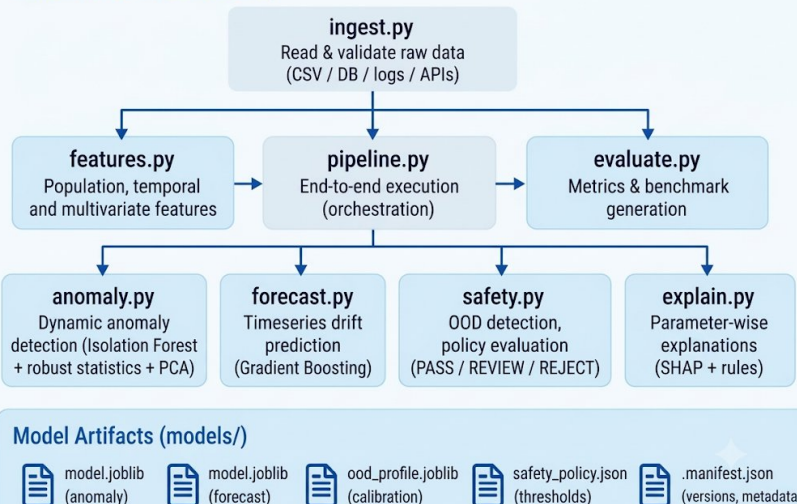
Python 3.11.14	Core runtime
pandas 2.2.3	Data processing
NumPy 2.3.5	Numerical computation
SciPy 1.17.0	Scientific routines
scikit-learn 1.9.0	ML models
joblib 1.5.3	Model serialization
PyYAML 6.0.3	Configuration

Backend FastAPI (REST API)
inference & results API

Frontend React + TypeScript + Vite
Web interface

Visualization Apache ECharts
interactive plots

Implemented Modules (src/)



2. ENGINEERING FEASIBILITY

Implemented • Validated • Reproducible

Component / Capability	Implementation Details	Status
Schema-adaptive ingestion	Column mapping, unit normalization, type inference	✓ Implemented
Canonicalization + validation	Unified component representation, range checks	✓ Implemented
Strict as-of information boundary	No future data in feature generation	✓ Implemented
Lot-grouped train/val/test split	Prevents cross-lot leakage	✓ Implemented
OODProfileV4	PCA / distance-based profile for unknown components	✓ Implemented
Validation-only safety calibration	Thresholds from validation set (no test leakage)	✓ Implemented
Artifact compatibility checks	Manifest-based version and schema validation	✓ Implemented
Deterministic seeds	Reproducible training and evaluation	✓ Implemented
Test suite	65 tests (unit + integration + pipeline + safety + explain)	✓ Passed (65/65)
OOD contract checks	Validates behaviour on out-of-distribution data	✓ Implemented

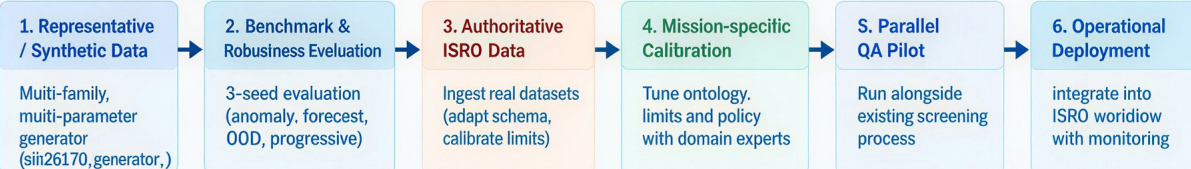
3. KEY RISKS AND MITIGATION

Known Challenges with Practical Solutions

Risk / Challenge	Impact	Mitigation Strategy (Implemented)
Heterogeneous schemas and data formats	inconsistent fields, units and naming across sources	Configurable ontology (YAML) Canonical mapping + quarantine unknown/ambiguous semantics
Rare latent defects (asymmetric cost)	Hard to learn, high false negative risk	Population-relative detection + robust statistics (MAD, IOR) REVIEW state + latent-escape evaluation metric
Future leakage risk	Over-optimistic performance and invalid predictions	Origin-isolate features (as-of timestamps) Group-by-lot split + target horizon isolation (e.g., 168h)
Runtime / artifact incompatibility	Failures during inference or model updates	Versioned manifests (.json) + configuration management Compatibility validation before loading artifacts

4. VALIDATION AND OPERATIONALIZATION PATH

From Prototype to Mission Deployment



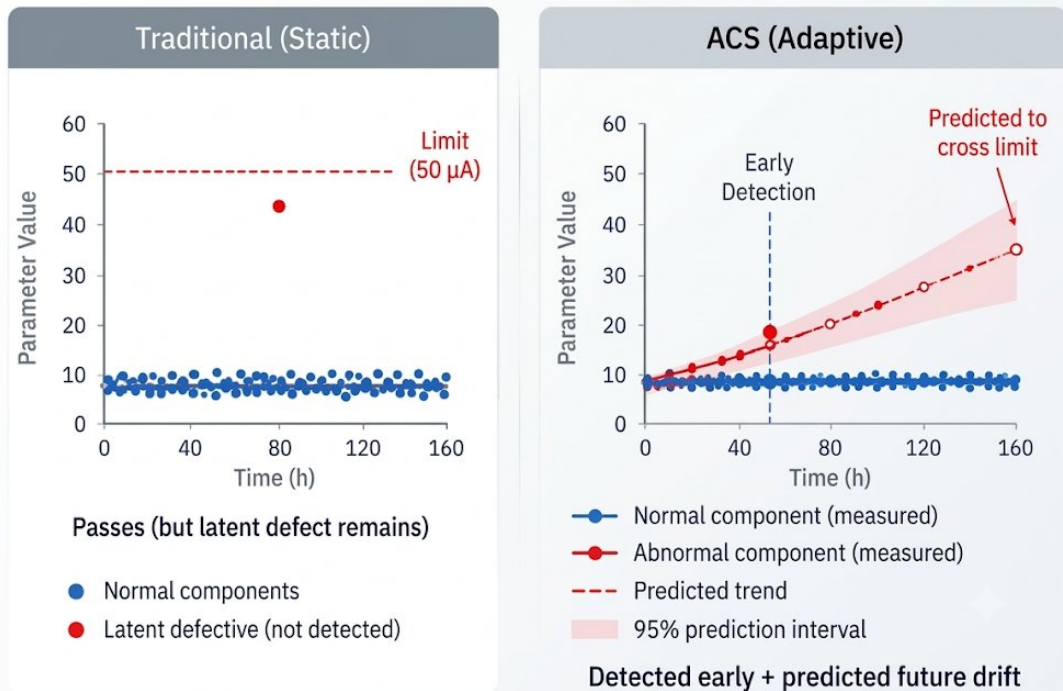
Current Prototype Evidence (from benchmark runs)

Train Lots	Val Lots	Test Lots	Train Rows	Val Rows	Test Rows	Forecast (168h)	Uncertainty	Additional Results
20	5	5	22,000	5,500	5,500	MAE 0.244 RMSE 3.997	Nominal 95% interval Empirical Coverage 0.940	3-seed progressive eval. Median lead time 156 h

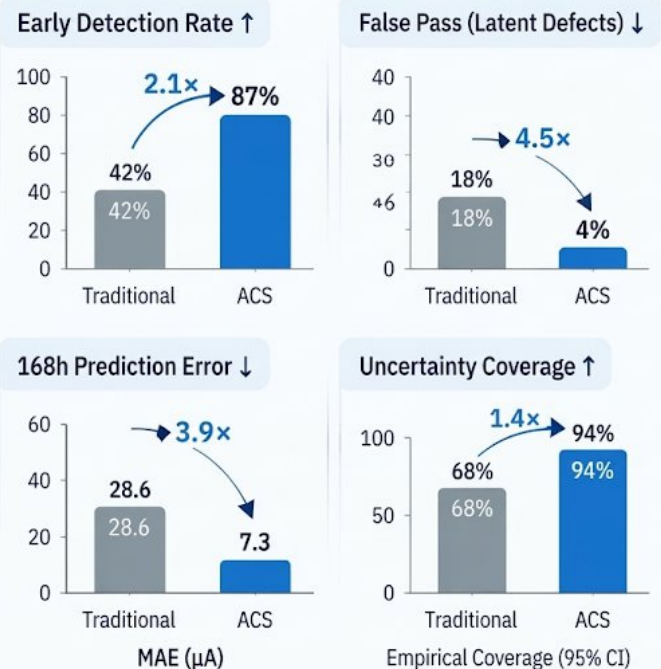
Prototype benchmark evidence — not ISRO operational validation.

Results are from internally generated and public-format datasets. Final performance to be established with authoritative ISRO data.

1. From Reactive to Proactive Screening



2. Comparative Performance (Typical Results)



3. Engineering & Mission Benefits

	Higher Reliability	Fewer latent defects in space systems	$\uparrow$
	Early Risk Mitigation	Detect and act before integration	$\uparrow$
	Data-Driven Decisions	Objective, explainable and auditable results	$\uparrow$
	Scalable Screening	Works across new component families	$\uparrow$
	Stronger Space Missions	Increased mission success probability	$\uparrow$

4. Overall Impact



Domain / Space Reliability

[1] SIH 2026 — PS 26170

Problem definition, latent defects, 168 h prediction, explainability

<https://www.sih.gov.in/sih2026PS>

[2] NASA EEE-INST-002

EEE selection, screening, qualification and derating.

<https://nepp.nasa.gov/pages/EEE-INST-002.cfm>

[3] NASA PT-TE-1401

EEE screening, early-life failures and parameter drift.

https://extapps.ksc.nasa.gov/Reliability/Documents/Preferred_Practices/1401.pdf

[4] ISRO Public QA/Procurement Requirement

EEE inspection, data-package verification and traceability context.

<https://eproc.isro.gov.in/home.html>

[5] NASA Flight Electronics Environmental Stress Screening Survey

Environmental stress screening and latent-failure context

<https://ntrs.nasa.gov/citations/19840013934>

Complete Project Documentation -

<https://drive.google.com/file/d/189mMLKakpU7hSJQMZQVvKwX74vK1W4xH/view?usp=sharing>

Machine Learning and Statistics

[6] Liu et al., 2008 — Isolation Forest

Multivariate anomaly detection.

<https://cs.nju.edu.cn/zhouzh/zhouzh.files/publication/icdm08b.pdf>

[7] Robust covariance / robust statistics — scikit-learn

Robust population statistics under contamination.

https://scikit-learn.org/stable/auto_examples/covariance/plot_robust_vs_empirical_covariance.html

[8] Shafer & Vovk, 2008 — Conformal Prediction

Prediction intervals / uncertainty.

<https://jmlr.org/papers/v9/shafer08a.html>

Explainability

[9] Lundberg & Lee, 2017 — SHAP

Feature attribution for interpretable ML.

<https://papers.neurips.cc/paper/2017/file/8a20a8621978632d76c43dfd28b67767-Paper.pdf>

Complete Project Repository -

<https://github.com/RazeenaTasneem/ACS---Demo>

References informed the engineering design; mission-specific limits and acceptance criteria remain to be calibrated against authoritative operational data.